

PHYS 121 — HW6 (Redesigned)

Student Name: _____.

NetID: _____.

Q1. Temperature of a pendulum (4+4 pts)

A pendulum has a rod of length L (negligible mass) that expands thermally and a compact bob. (a)

Using $T \approx 2\pi \sqrt{L/g}$ and $dL = \alpha L dT$, show the fractional change $dT_{\text{period}} / T = (1/2) \alpha dT$. (b)

A brass pendulum clock is correct at T_{cal} . In a room at T_{room} ($T_{\text{room}} > T_{\text{cal}}$), does it run fast or slow?

Compute the error in seconds/day in terms of α and $\Delta T = T_{\text{room}} - T_{\text{cal}}$. Required drawing:

pendulum sketch with L and thermal expansion δL labeled.

Q2. Estimate Earth's temperature (2+3+3+1 pts)

As the very first rudiment of climatology, estimate the temperature of Earth. Assume it is a perfect sphere and its temperature is uniform. Ignore the greenhouse effect. Thermal radiation from the Sun has an intensity (the “solar constant” S) of about 1370 W/m^2 at the radius of Earth's orbit. (a) Assuming the Sun's rays are parallel, what area must S be multiplied by to get the total radiation intercepted by Earth? It will be easiest to answer in terms of Earth's radius, R . (b) Assume that Earth reflects about 30% of the solar energy it intercepts. In other words, Earth has an albedo with a value of $A = 0.3$. In terms of S , A , and R , what is the rate at which Earth absorbs energy from the Sun? (c) Find the temperature at which Earth radiates energy at the same rate. Assume that at the infrared wavelengths where it radiates, the emissivity e is 1. Does your result show that the greenhouse effect is important? (d) How does your answer depend on the area of the Earth?

Q3. Unreasonable results? H₂S speeds (3+5 pts)

(a) Find the average speed of hydrogen sulfide (molar mass 31.4 g/mol), H_2S , at $T = 250$ K. (b) Why is this result less reliable than for ideal monoatomic or nearly ideal gases like neon? Discuss non-ideality and molecular interactions.

Q4. Airtight water dispenser (10 pts)

A rigid, airtight dispenser has a horizontal $25\text{ cm} \times 10\text{ cm}$ cross-section and a height of 20 cm . A tap at the bottom opens to the atmosphere. Initially: tap closed, water level is 3.5 cm below the top, with **air** above at 1.00 atm . Temperature constant; water density 1000 kg/m^3 . When the tap is opened, water flows until the gauge pressure at the bottom is zero. (a) Write $P_{\text{bottom}} = P_{\text{air}} + \rho g h$. For gauge = 0, set $P_{\text{bottom}} = P_{\text{atm}} = 1.00\text{ atm}$. (b) With isothermal air, use $P_{\text{air}} V_{\text{air}} = \text{constant}$ to relate air pressure to water height. (c) Solve for the final water height and report the volume of water that flows out. Required drawing: side-view geometry labeling water height h and air pocket size.

Q5. Adiabatic process (3+3+3+3 pts)

One mole of an ideal monatomic gas occupies a volume of $V_1 = 1.0 \times 10^{-2} \text{ m}^3$ at a pressure of $2.0 \times 10^5 \text{ N/m}^2$. (a) What is the temperature of the gas? (b) The gas undergoes a quasi-static adiabatic compression until its volume is decreased to $V_2 = 5.0 \times 10^{-3} \text{ m}^3$. What is the new gas temperature? (c) How much work is done on the gas during the compression? (d) What is the change in the internal energy of the gas? Required drawing: P–V sketch with adiabatic curve $V_1 \rightarrow V_2$.

Q6. Inelastic collision heating (5+5 pts)

A bullet of mass 10 g is traveling horizontally at 200 m/s when it strikes and embeds in a pendulum bob of mass 2.0 kg. (a) How much mechanical energy is dissipated in the collision? (b) Assuming that C_V for the bob plus bullet is $3R$, calculate the temperature increase of the system due to the collision. Take the molecular mass of the system to be 200 g/mol.

Q7. A Star is Born (3+3+3+2 pts)

A spherical hydrogen cloud of initial radius R_i and uniform density ρ_i compresses. At first, it stays at constant T_i until it becomes opaque at density ρ_1 . After that, the collapse is adiabatic with $\gamma = 7/5$. (a) Use mass conservation to find r_1 when $\rho = \rho_1$. (b) In the adiabatic phase, derive how T scales with radius and obtain r_2 when the cloud reaches temperature $T_2 > T_1$. (c) With $R_i = 3.0 \times 10^{15}$ m, $\rho_i = 2.0 \times 10^{-14}$ g/m³, $\rho_1 = 1.0 \times 10^{-7}$ g/m³, compute r_1 numerically. (d) If the gas glows visibly at $T_2 = 2000$ K, find r_2 in terms of your T_1 . Then evaluate numerically once T_1 is specified. (e) We can't go much further with this model, because the pressure of the cloud will not be uniform if the cloud gets too small. The physics doesn't get harder| it is still Pascal's equation, the ideal gas equation, and Newton's law of gravitation| but they need to be connected with differential equations. So, that's all, folks! [Well, this is a comment, not a question...]

Q8. A process: piston + mass, adiabatic vs isothermal (3+4+4+3 pts)

A vertical cylinder contains 1 mol He at $T_1 = 27^\circ\text{C}$ and $P_1 = 100.0\text{ kPa}$. Piston area $A = 0.0010\text{ m}^2$. The cylinder sits in a 27°C water bath. A 2.0 kg mass is gently placed on the piston and then released from rest. The piston initially oscillates. (a) Minimum volume during the first dip: use energy conservation (gravitational drop converts to adiabatic internal-energy rise). (b) After oscillations damp out (still adiabatic overall), find the equilibrium volume via force balance. (c) Eventually, the gas re-equilibrates with the bath (27°C). Find the final equilibrium volume. (d) Compute the net heat absorbed or released by the gas over the entire process. Required drawing: PV sketch of dips and final states; side view with forces on piston.

Q9. A cycle with N₂ (3+3+2+3 pts)

Two moles of N₂ ($\gamma \approx 7/5$) begin at $T = 300$ K and volume $V_a = 0.0100$ m³. The gas is compressed adiabatically and reversibly to 5 L, then the piston is locked and the gas cools at constant volume back to 300 K. Finally, it expands isothermally at 300 K back to the original volume, closing the cycle. (a) Compute the net work over the cycle. Break it into adiabatic, isochoric (zero work), and isothermal parts. (b) Compute the net heat into/out of the gas. (c) Compute ΔU over the entire cycle. (d) Compute the entropy change ΔS of the gas over the cycle. Required drawing: PV diagram with labeled paths and states.

Q10. Carnot engine with N₂ (4+4+3+3+1+3 pts)

An ideal reversible Carnot engine employs 1 mol of nitrogen. $T_h = 400$ K, $T_c = 300$ K. Initial volume at a is $V_a = 6.0$ L; at c it is $V_c = 18.0$ L. (a) Find V_b where the cycle switches from isothermal expansion to adiabatic expansion. (b) Find V_d where adiabatic compression ends. (c) Heat input on $a \rightarrow b$, Q_{in} . (d) Heat out on $c \rightarrow d$, Q_{out} . (e) Engine efficiency η . (f) Entropy change per gram of the working substance during ab and bc. Required drawing: Carnot loop in P–V, with states a, b, c, d.

Bonus (0–2 pts): Greenhouse effect

Briefly describe what the greenhouse effect is and your 2-3 suggestions on how to make Earth a better place to live. [less than 200 words]

Assistance notes: